Adhvaith Vemula

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.99
Narayana Junior College	Class XII (SSC)	2024	98.40%
Narayana High School	Class X (SSC)	2022	9.70%

PROJECTS

Chess Engine (C++)

- Engineered a **UCI-compliant chess engine** in **C++** with **bitboard representation** and full rule coverage, **castling**, **en passant**, **promotion**, **FEN parsing**.
- Implemented **negamax with alpha-beta pruning**, **quiescence search**, **iterative deepening**, and **MVV-LVA ordering**, backed by **Zobrist hashing** and a **transposition table**.
- Built a **UCI protocol handler** and engine loop, validated via extensive **perft testing** for move-generation accuracy and **make/unmake** correctness.

2D Multiplayer Game (C# / .NET 8)

- Developed a **real-time multiplayer 2D game** in **C# 12/.NET 8** with a **client-server architecture (Shared/Server/Client)** using raw **WebSockets** and **Raylib-cs** for rendering.
- Implemented **client-side prediction**, **server reconciliation**, and **entity interpolation** for responsive gameplay under network latency while maintaining **server-authoritative state**.
- Built **server-authoritative hit detection** to prevent client-side cheating, with **polymorphic JSON serialization** for structured client-server message passing.

Hospital Management System (ASP.NET Core / React / PostgreSQL)

- Architected a **production-grade, full-stack hospital management system** ,a **web dashboard (React 19 + TypeScript)** and **cross-platform mobile app (React Native/Expo, iOS & Android)** powered by a **single unified ASP.NET Core REST API** with **PostgreSQL** and **Entity Framework Core**.
- Designed a **secure, hierarchical RBAC system** using **JWT authentication** across **4 roles**, enforcing **per-doctor data scoping** server-side via **token claims** to prevent unauthorized cross-doctor access.
- Built end-to-end hospital workflows , **patient registration**, **itemized billing with audit trails**, and a **revenue analytics engine** with **daily/monthly/yearly, per-doctor breakdowns** ,plus a mobile client with **offline-first caching** and **full feature parity** with the web app.

xv6 Operating System Enhancements (C, xv6, RISC-V, QEMU)

- Extended the **xv6 RISC-V kernel** with **custom system calls**, **process relationship APIs**, and **syscall accounting**, working directly with **kernel internals** and **user-kernel boundary design**.
- Designed a **System-Call-Aware MLFQ scheduler** with **dynamic priority adjustment**, **time-slice management**, **promotion/demotion**, and **starvation prevention**.
- Built a **virtual memory subsystem** with **lazy allocation**, **LRU/CLOCK page replacement**, and **swapping** for transparent **swap-in/swap-out** under memory pressure.

Molecular Dynamics Simulator for Atomistic Material Analysis | Python, NumPy

- Engineered a **Molecular Dynamics engine** from scratch with **Lennard-Jones forces**, **Periodic Boundary Conditions**, **Minimum Image Convention**, and **FCC lattice generation**.
- Implemented **Velocity Verlet integration**, **Maxwell-Boltzmann initialization**, and **thermostats**, optimized via **Verlet Neighbor Lists** and **NumPy vectorization**.
- Built an **equilibration-to-production pipeline** with **RDF/MSD analysis** to characterize **melting behavior** and **transport properties**.

RELEVANT COURSES

***Computer Science**:* Introduction to Programming, Data Structures and Applications, Operating Systems I & II

***Scientific Computing**:* Computer Aided Numerical Methods I & II, Fundamentals of Scientific Computing, Computational Methods in Material Science

***Mathematics**:* Introduction to Algebra, Introduction to Statistics, Linear Algebra, Transform Techniques

SKILLS

***Programming Languages**:* C, C++, Python, C#, JavaScript, TypeScript

***Web Development**:* HTML, CSS, ASP.NET Core, WebSockets

***Frameworks & Technologies**:* .NET

***Databases**:* PostgreSQL

POSITIONS OF RESPONSIBILITY

Head, Shuffle Crew (Dance Club)

Coordinator, Public Relations & Outreach -- Finance and Consulting Club

Coordinator, Operations and Logistics -- TEDx

Coordinator, Operations and Logistics -- International Relations Cell

EXTRACURRICULAR

- Represented IIT Hyderabad in the Inter-IIT Cultural Meet as part of the dance team, competing against contingents from other IITs
- Active sports enthusiast -- regularly play cricket, badminton, and table tennis